

The empirical recurrence for OEIS A230472, recovered from its tail

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Abstract

OEIS A230472 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 140$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 59$. Its last coefficient is $c_{57} = 24461180928 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A230472 is “Number of $4 \times n$ 0..2 arrays $x(i,j)$ with each element horizontally, diagonally or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 3$, and upper left element zero.”. It has offset 1 and begins

0, 8, 1384, 76518, 3062214, 131421154, 5660619846, 245148922672, ...

The entry states:

Empirical recurrence of order 57 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 05:51:46, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 140$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 140$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 280$ exact terms from $n = 2$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	38	c_{16}	−1040934207014	c_{31}	−8122278392726950	c_{46}	−4247286832559040
c_2	249	c_{17}	2712455033187	c_{32}	7736562024279662	c_{47}	1416119384082432
c_3	−1631	c_{18}	−4839259238799	c_{33}	−9330706625667558	c_{48}	588775181364224
c_4	25787	c_{19}	5109746288251	c_{34}	11785101328780676	c_{49}	−656580468634112
c_5	218698	c_{20}	34176085322110	c_{35}	−9807835993486308	c_{50}	164751660097536
c_6	−2654095	c_{21}	−114617101391511	c_{36}	65288385165842	c_{51}	123593944840192
c_7	−16902770	c_{22}	238801759302198	c_{37}	6256150705032020	c_{52}	−79829506486272
c_8	−45411962	c_{23}	−769544447946036	c_{38}	−823163325660784	c_{53}	−794670391296
c_9	−698912465	c_{24}	1260717703623858	c_{39}	−7037934437852200	c_{54}	9601803288576
c_{10}	1770363413	c_{25}	−2373652514428879	c_{40}	1999185851141080	c_{55}	−1947088060416
c_{11}	−5621060592	c_{26}	3453018182040042	c_{41}	9108142811257536	c_{56}	−11636047872
c_{12}	18988394069	c_{27}	−3884629521528852	c_{42}	−6224803677155848	c_{57}	24461180928
c_{13}	14487968840	c_{28}	5422774684373062	c_{43}	−7731425190178448		
c_{14}	−38573063652	c_{29}	−5226970450954395	c_{44}	10014857398504192		
c_{15}	467459546009	c_{30}	6692732094408203	c_{45}	30734807127264		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 59$, and it is the recurrence OEIS A230472 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{57-1} - \dots - c_{57}$. Its constant term is $-c_{57} = -24461180928$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 24461180928.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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